

WASP-49b

R-1257

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_200214 · created 2026-07-31T12:20:29+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458893.6473 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.041
Transit depth (Rp/Rs)^2	2.0 +/- 1.15 [%]
Orbital Inclination (inc)	85.37 +/- 2.88
Airmass coefficient 1 (a1)	0.87 +/- 0.58
Airmass coefficient 2 (a2)	0.07 +/- 1.56
Scatter in the residuals of the lightcurve fit is	63.59 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.25
Transit Duration (day)	0.082 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.041 vs 0.116 (deviation 0.61σ)
Duration fit vs published	0.082 d vs 0.09 d (deviation 0.27σ)
Mid-time O–C	5.1 min
Depth SNR	3.4
Residual scatter	63.59 %

Light curve

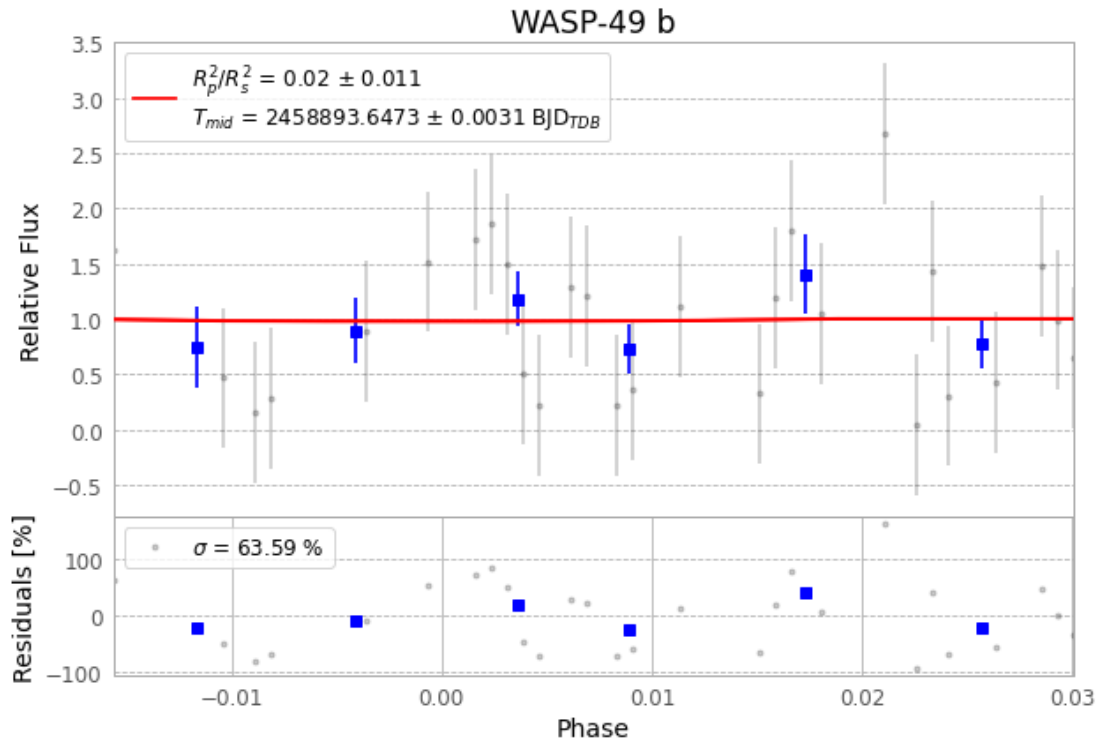


Figure 1 — FinalLightCurve_WASP-49 b_2020-02-13.png (SHA-256 39788c8793640b83...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [208, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aacd4373ff78d4f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-49200214022412.FITS → WASP-49200214053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-200214.log (0fa3f398584f...), FinalLightCurve_WASP-49 b_2020-02-13.png (39788c879364...), FinalLightCurve_WASP-49 b_2020-02-13.csv (2b838baf9d2c...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:20Z.